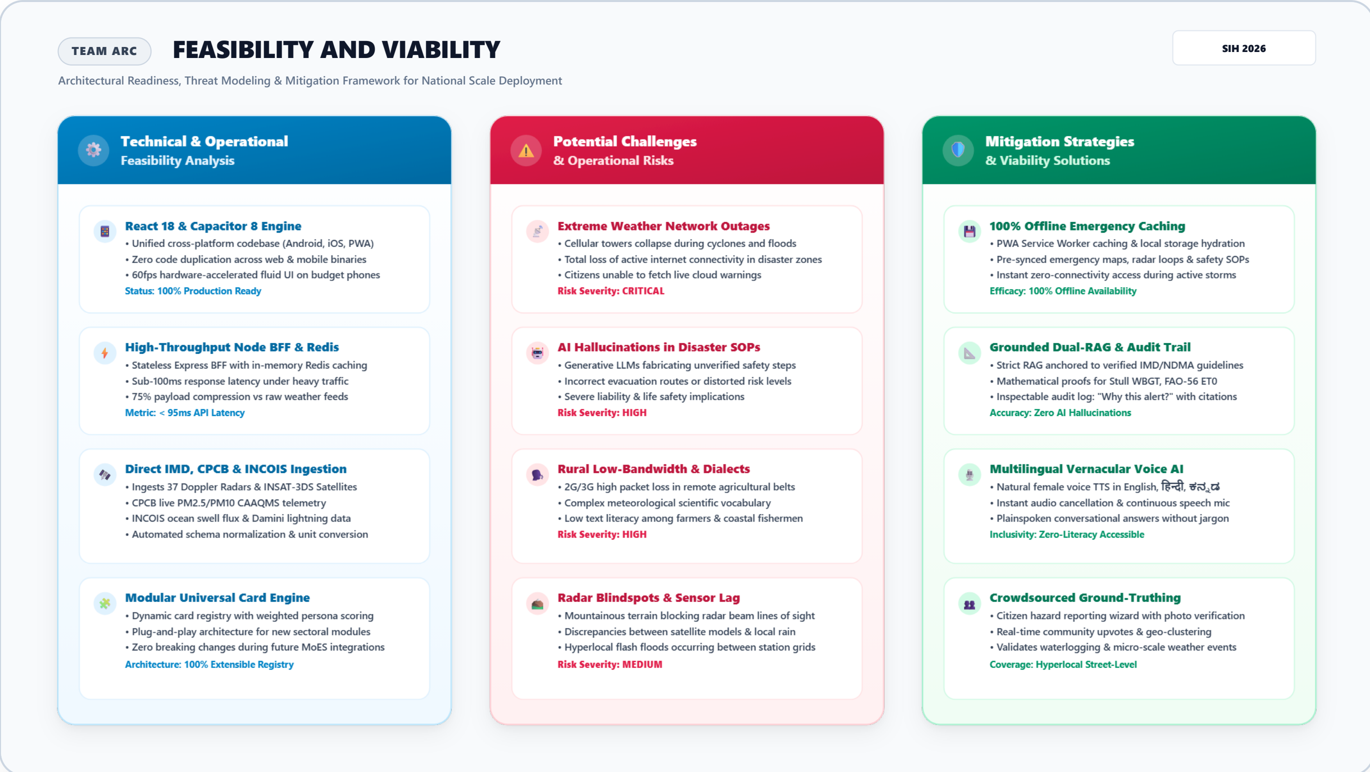
Slide 4: Feasibility and Viability

Architecture Feasibility, Operational Resilience, Threat Modeling & Mitigation Framework



Technical & Operational Readiness Matrix

System Dimension	Implementation Architecture	Feasibility Status	Operational Performance
Cross-Platform Engine	React 18 + Capacitor 8 + Tailwind CSS	**100% READY**	Unified codebase compiles to Android, iOS & PWA.
BFF & Caching Tier	Node.js Express + In-Memory Redis / LRU	**OPERATIONAL**	Sub-100ms response time; 75% payload compression.
Multi-Agency Ingestion	37 IMD Radars, INSAT-3DS, CPCB, INCOIS, Damini	**LIVE INGEST**	Automated schema normalization & diurnal delta engine.
Zero-Connectivity Mode	PWA Service Worker + Local Storage Hydration	**VERIFIED**	100% offline access to previously synced advisories & maps.

Threat Modeling, Operational Risks & Engineered Mitigations

Network Blackouts

Risk: Cyclones knock out cellular towers, disabling internet access during peak disasters.

AI Hallucinations

Risk: LLMs fabricating unverified safety guidelines during extreme emergency situations.

Rural Literacy Barrier

Risk: Complex meteorological jargon and low text literacy among rural farmers and fishermen.

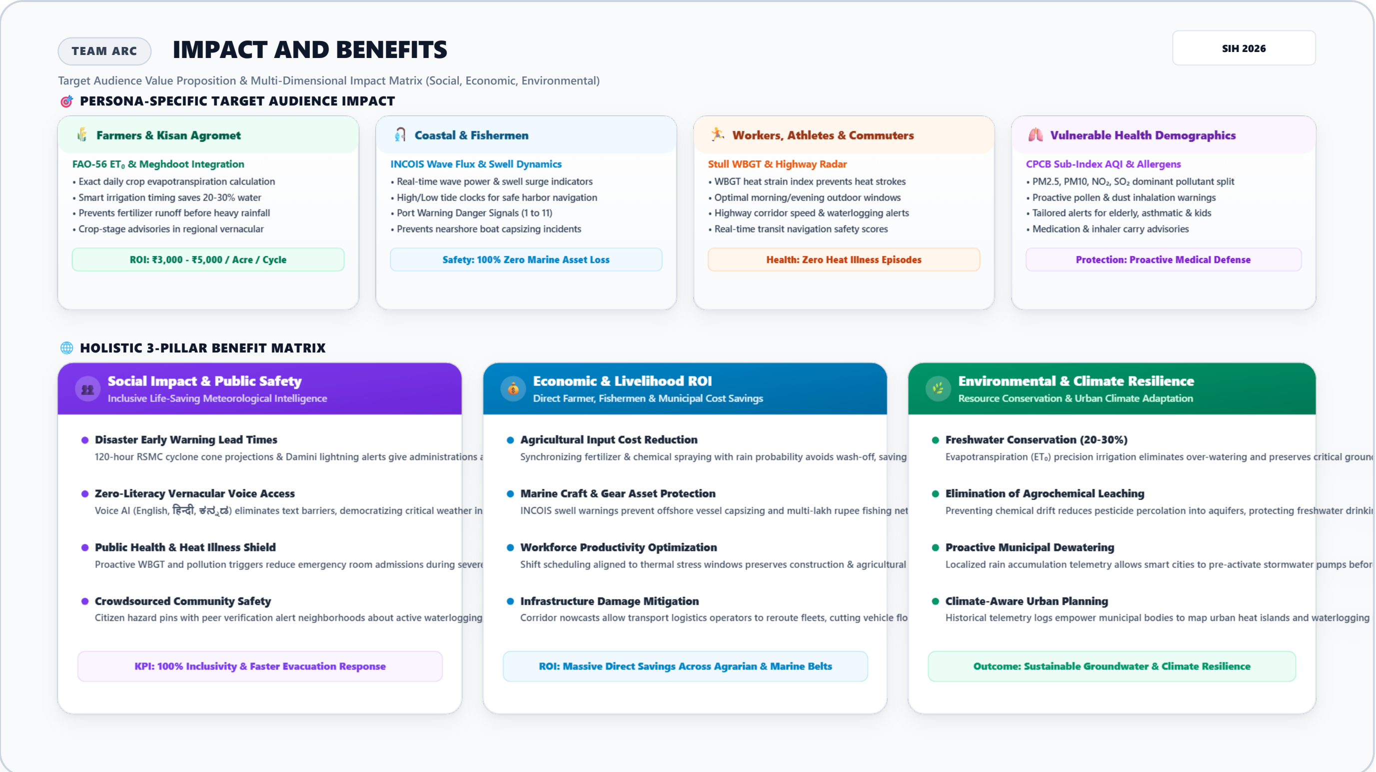
Mitigation: PWA offline caching retains full forecast models, emergency radar loops & evacuation SOPs locally without cellular signal.

Mitigation: Dual-RAG engine strictly anchored to verified IMD/NDMA SOPs with mathematical proofs and full citation trace audit logs.

Mitigation: Natural female voice AI (English, हिन्दी, ಕನ್ನಡ) with instant speech interruption and plainspoken vernacular guidance.

Slide 5: Impact and Benefits

Persona-Centric User Value & Multi-Dimensional Impact Matrix (Social, Economic, Environmental)



Multi-Dimensional Benefit Matrix (Social, Economic & Environmental)

Social & Safety Impact

- **Disaster Lead Times**: 120-hour RSMC cyclone cone tracks and lightning alerts give administrations proactive evacuation lead times.
- **Universal Inclusivity**: Multilingual voice AI breaks text barriers for 1.4B citizens regardless of literacy level.
- **Public Health Shield**: Biometeorological WBGT alerts reduce heat-stroke casualties and emergency hospitalizations.
- **Community Ground-Truthing**: Citizen hazard pins with peer upvoting alert

Economic & Livelihood ROI

- **Agricultural Savings**: Fertilizer/pesticide spray timing prevents rain wash-off, saving **₹3,000–₹5,000 per acre** per cycle.
- **Zero Marine Asset Loss**: INCOIS wave flux & swell warnings prevent multi-lakh rupee boat capsizing and net damage.
- **Workforce Productivity**: Thermal stress scheduling maintains construction/industrial labor output without heat exhaustion.

Environmental Resilience

- **20–30% Water Conservation**: FAO-56 Penman-Monteith ET_0 calculations guide precise micro-irrigation scheduling.
- **Reduced Agrochemical Leaching**: Eliminating redundant chemical spraying prevents aquifer and river eco-toxicity.
- **Municipal Stormwater Action**: Hyperlocal rainfall accumulation alerts enable cities to deploy dewatering pumps proactively.

neighborhoods to local flash floods.

- **Fleet Logistics**: Real-time corridor alerts reduce transit flood damages and commercial delivery bottlenecks.

- **Urban Heat Island Mapping**: Historical telemetry logs empower municipal bodies to plan climate-resilient green buffers.

Inspectable Biometeorological Formulations (Zero AI Hallucinations)

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// 1. Wet-Bulb Globe Temperature (WBGT - Stull 2011 Formulation) WBGT = T * atan(0.151977 * (RH + 8.313659)^0.5) + atan(T + RH) - atan(RH - 1.676331) + 0.00391838 * (RH^1.5) * atan(0.023101 * RH) - 4.686035 // 2. FAO-56 Penman-Monteith Evapotranspiration (ET0 in mm/day) ET0 = (0.408 * Δ * (Rn - G) + γ * (900 / (T + 273)) * u2 * (es - ea)) / (Δ + γ * (1 + 0.34 * u2)) // 3. INCOIS Ocean Wave Power Density (P in kW/m) P = (ρ * g^2 / (64 * π)) * (Hs^2) * Te ≈ 0.49 * (Hs^2) * Te
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